P. R. Pote Patil College of Engineering and Management, Amravati

Title of Project

Public Health Data Analysis – Diabetes Dataset

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CERTIFICATE

This is to certify that following group of students of have successfully completed the AI&DS Lab project titled **"Public Health Data Analysis – Diabetes Dataset "** for the requirements of the Bachelors of Technology (B.Tech) program in AI&DS Laboratory during the academic year 2025–26.

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Abstract

Public Health Data Analysis – Diabetes Dataset is a data-driven analytical project developed to study and understand diabetes trends using real-world healthcare data. The project focuses on analyzing medical and lifestyle parameters to identify patterns, risk factors, and relationships that contribute to diabetes prediction and prevention.

The application is designed to be simple, informative, and accessible for students, beginners, and researchers interested in public health analytics. By combining data science techniques with healthcare information, the project improves analytical skills and promotes awareness about diabetes risk factors and early detection.

Overall, the Public **Health Data Analysis – Diabetes Dataset** project provides an effective platform for understanding diabetes through data analysis, making healthcare information more visual, insightful, and easy to interpret.

ACKNOWLEDGEMENT

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Chapter 1

INTRODUCTION

1.1 Objective

1. To load and understand the dataset
2. To perform Exploratory Data Analysis (EDA)
3. To handle missing and incorrect values
4. To remove outliers using IQR method
5. To normalize the dataset
6. To visualize important features
7. To identify patterns related to diabetes

1.2 Introduction

Public health data plays an important role in understanding diseases and improving healthcare decisions. Diabetes is one of the most common diseases worldwide. This project uses the Pima Indians Diabetes Dataset to analyze health-related factors such as glucose level, BMI, insulin, and blood pressure.

The aim is to explore the dataset, clean the data, and identify patterns that can help in understanding diabetes.

This project focuses on the analysis of public health data using the Pima Indians Diabetes Dataset. The main objective is to understand patterns related to diabetes and improve data quality through preprocessing techniques. The dataset is cleaned by handling missing values and removing outliers.

Further, normalization is applied to scale the data. Various visualizations such as histograms, heatmaps, countplots, and boxplots are used to analyze relationships between variables. The project helps in identifying key factors affecting diabetes and demonstrates a complete data analysis workflow using Python libraries.

Public health data helps in understanding diseases and improving healthcare decisions. Diabetes is a major global health issue affecting millions. This project uses the Pima Indians Diabetes Dataset.

Goal: Analyze health indicators to identify diabetes patterns.

Chapter 2

PROPOSED WORK

2.1 Basic Idea

The project **Public Health Data Analysis – Diabetes Dataset** aims to analyze healthcare data to understand patterns, risk factors, and trends related to diabetes. The system focuses on applying data science techniques to extract meaningful insights from medical datasets, helping users understand how different health parameters contribute to diabetes prediction.

Instead of traditional theoretical study, this project adopts a data-driven analytical approach where real-world health data is processed, cleaned, visualized, and interpreted. The system allows users to explore relationships between medical attributes such as glucose level, blood pressure, body mass index (BMI), insulin level, age, and other health indicators.

The main objective of this project is to transform raw healthcare data into useful information through preprocessing and Exploratory Data Analysis (EDA). This helps simplify complex public health data and supports better understanding of diabetes risk factors and preventive healthcare measures.

Key Features

The software is designed to make public health data analysis simple, informative, and visually understandable through analytical techniques.

1. Healthcare Data Analysis
2. Analyzes diabetes dataset to identify patterns and important health indicators.
3. Data Cleaning and Preprocessing
4. Handles missing values, incorrect entries, and prepares data for analysis.
5. Exploratory Data Analysis (EDA)
6. Uses statistical and visual methods to explore relationships between variables.
7. Data Visualization
8. Graphs and plots are used to represent trends and correlations clearly.
9. Feature Understanding
10. Studies medical parameters affecting diabetes occurrence.
11. Improved Data Quality
12. Removes noise and outliers for accurate analytical results.
13. User-Friendly Analytical Approach
14. Designed for students and beginners to easily understand healthcare analytics.

Processed Work (Data Preprocessing & Exploratory Data Analysis)

The diabetes dataset initially contained 768 records and 9 features. The following preprocessing and analysis steps were performed:

✓ **Data Inspection**
- Checked dataset shape, column names, and data types
- Displayed initial rows to understand dataset structure

✓ **Handling Missing Values**
- Certain features contained missing or zero values
- Invalid values were replaced with appropriate statistical measures
- Dataset was cleaned to ensure no missing values remained

✓ **Outlier Detection and Removal**
- Boxplots were used to identify abnormal or extreme values
- Outliers were removed to improve data consistency
- Dataset size was reduced to 332 rows after cleaning

✓ **Data Normalization**
- All numerical features were scaled between 0 and 1
- Improved consistency and performance for further analysis

✓ **Data Visualization**
- Histograms plotted to analyze feature distributions
- Boxplots used for detecting data spread and outliers
- Countplot visualized class imbalance in diabetes cases
- Correlation heatmap generated to understand relationships between variables

2.2. Methodology

The following methodology was adopted for the Public Health Data Analysis – Diabetes Dataset project:

✓ **Step 1: Data Collection**

- The diabetes dataset was collected from an online source in CSV file format.
- The dataset contains medical attributes related to diabetes prediction such as glucose level, blood pressure, BMI, age, and insulin level.

✓ **Step 2: Exploratory Data Analysis (EDA)**

- Dataset shape, column names, and summary statistics were examined.
- Initial analysis was performed to understand data distribution and feature characteristics.
- Missing and zero values present in the dataset were identified.

✓ **Step 3: Data Cleaning**

- Zero or invalid values in specific medical attributes were replaced using the median value to maintain data consistency.
- Missing values were properly handled to ensure a clean and usable dataset.

✓ **Step 4: Outlier Detection & Removal**

- Extreme values were detected using the Interquartile Range (IQR) method.
- Outliers were removed to improve accuracy and reliability of analysis results.

✓ **Step 5: Data Normalization**

- Min–Max Scaling technique was applied to normalize all numerical features.
- Data values were scaled between 0 and 1, ensuring uniform feature range and improved analytical performance

✓ **Step 6: Data Visualization**

- Histograms were plotted to analyze feature distributions.
- Boxplots were used to observe spread and detect remaining outliers.
- Countplot was used to visualize class distribution of diabetic and non-diabetic cases.
- Correlation Heatmap was generated to identify relationships among different health attributes.

Figure 2.3.1:Methodology

Chapter 3

RESULT

The dataset was successfully cleaned and processed. Missing values were handled, and outliers were removed. The normalized dataset helped in better analysis. Visualizations showed that features like glucose and BMI have a strong relationship with diabetes. The final cleaned dataset was saved for further use. The project successfully demonstrates how data analysis can be used in healthcare

- **Result Screenshots**

```
--------------------------------------------------
STEP 1 — Loading Dataset
--------------------------------------------------
Dataset Shape: (768, 9)
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1

--------------------------------------------------
STEP 2 — Basic EDA
--------------------------------------------------

Columns:
Index(['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin'
       'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome'],
      dtype='object')

Missing Values:
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64

Summary:
       Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin  \
count   768.000000  768.000000     768.000000     768.000000  768.000000
mean      3.845052  120.894531      69.105469      20.536458   79.799479
std       3.369578   31.972618      19.355807      15.952218  115.244002
min       0.000000    0.000000       0.000000       0.000000    0.000000
25%       1.000000   99.000000      62.000000       0.000000    0.000000
50%       3.000000  117.000000      72.000000      23.000000   30.500000
75%       6.000000  140.250000      80.000000      32.000000  127.250000
max      17.000000  199.000000     122.000000      99.000000  846.000000

              BMI  DiabetesPedigreeFunction         Age     Outcome
count  768.000000                768.000000  768.000000  768.000000
mean    31.992578                  0.471876   33.240885    0.348958
std      7.884160                  0.331329   11.760232    0.476951
min      0.000000                  0.078000   21.000000    0.000000
25%     27.300000                  0.243750   24.000000    0.000000
50%     32.000000                  0.372500   29.000000    0.000000
75%     36.600000                  0.626250   41.000000    1.000000
max     67.100000                  2.420000   81.000000    1.000000
```

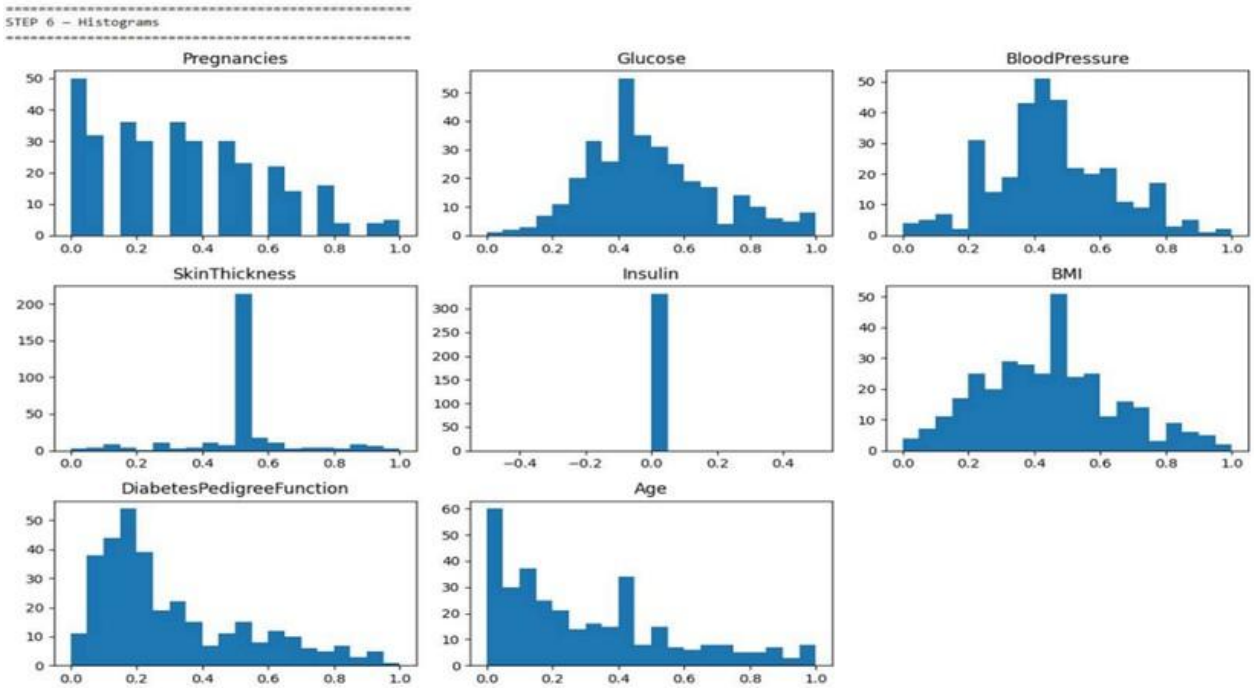
Figure 3.2.1: Index Page

```
Outcome Count:
Outcome
0    500
1    268
Name: count, dtype: int64

==================================================
STEP 3 — Handling Missing Values
==================================================
Missing values after filling:
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64

==================================================
STEP 4 — Removing Outliers
==================================================
Rows after outlier removal: 332

==================================================
STEP 5 — Normalization
==================================================
   Pregnancies   Glucose  BloodPressure  SkinThickness  Insulin       BMI
0     0.461538  0.684211       0.428571       0.740741      0.0  0.518519
1     0.076923  0.269737       0.321429       0.518519      0.0  0.282828
2     0.615385  0.914474       0.285714       0.518519      0.0  0.171717
3     0.384615  0.473684       0.464286       0.518519      0.0  0.249158
4     0.769231  0.467105       0.428571       0.518519      0.0  0.575758

   DiabetesPedigreeFunction       Age  Outcome
0                  0.581568  0.617021        1
1                  0.289195  0.212766        0
2                  0.629237  0.234043        1
3                  0.130297  0.191489        0
4                  0.059322  0.170213        0
```

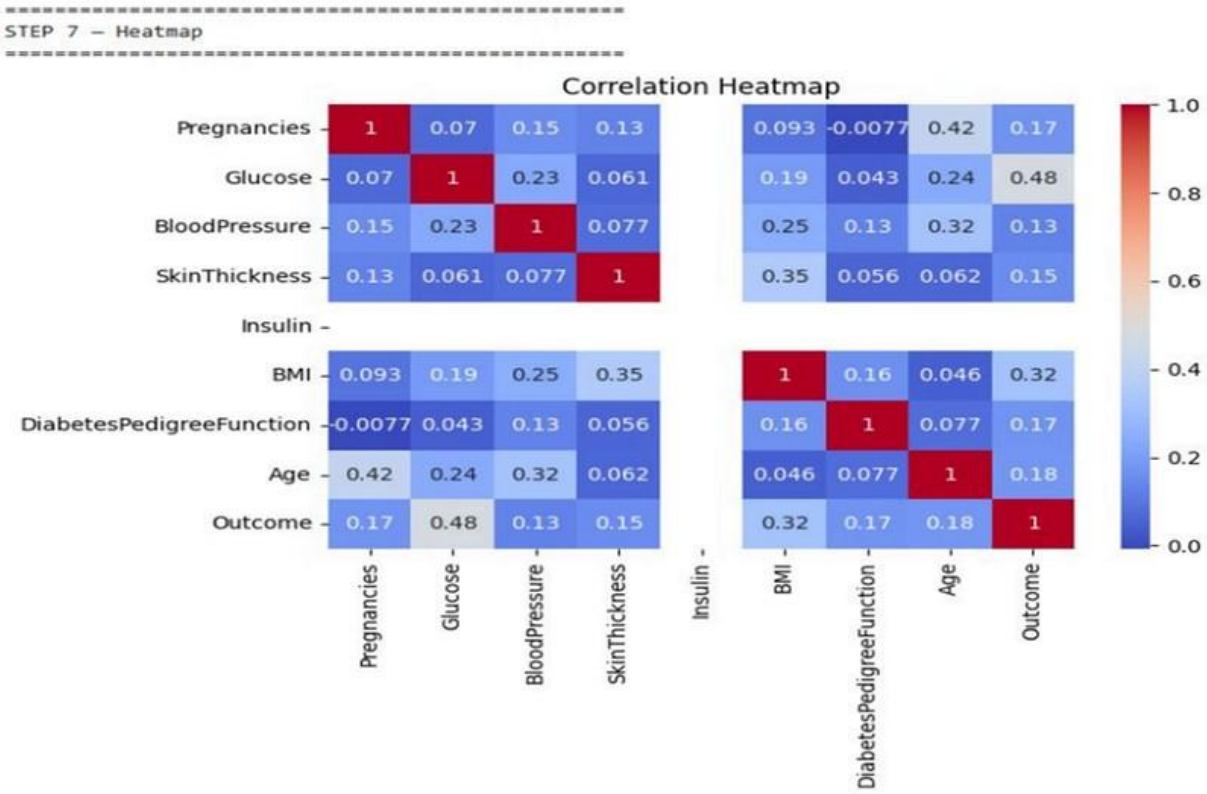
```
==================================================
STEP 10 — Saving File
==================================================
File saved successfully!

==================================================
PROJECT COMPLETED
==================================================
Final Rows: 332
Diabetic Patients: 115
Non-Diabetic Patients: 217

Saved files inside outputs folder:
1. Histograms
2. Heatmap
3. Countplot
4. Boxplots
5. Cleaned CSV
```

- **Graphs**

  - **Histograms show the distribution of features.**

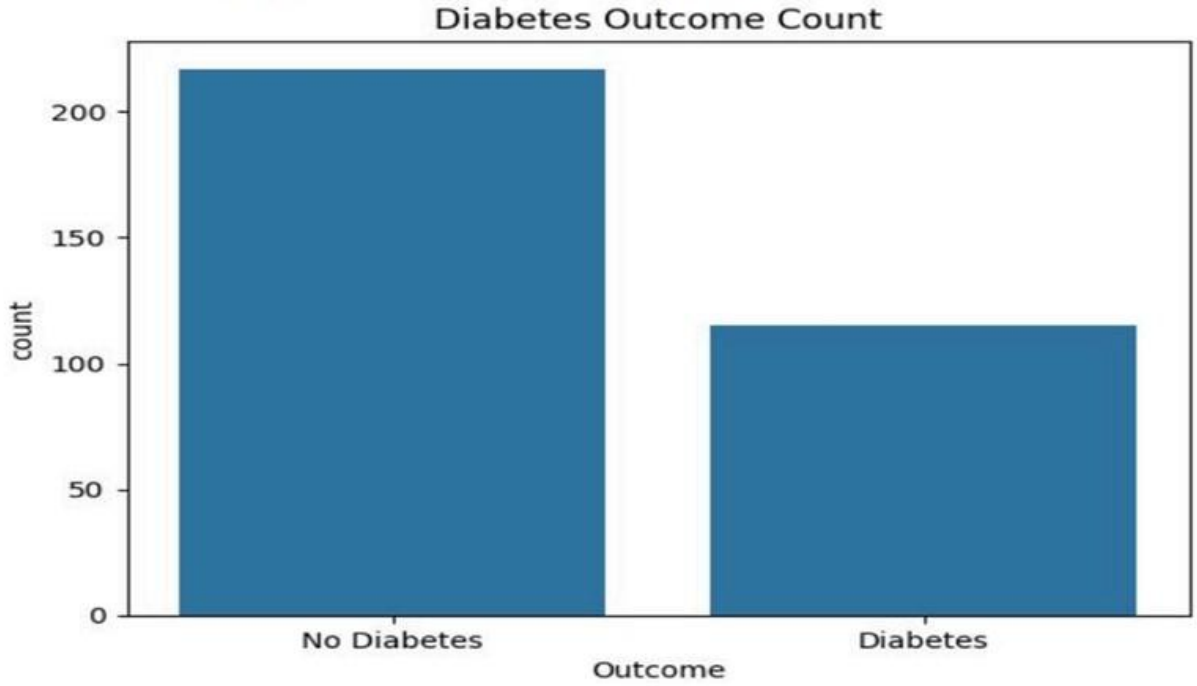


  - **Heatmap shows correlation between variables.**



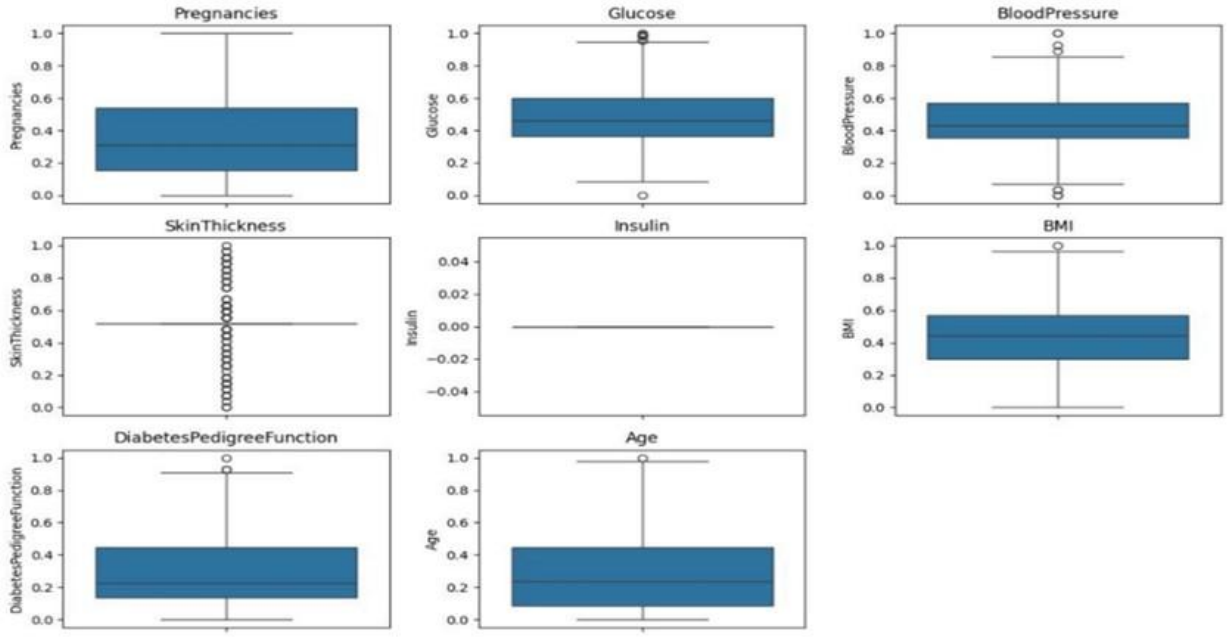
•**Countplot shows diabetic and non-diabetic cases.**

```
==================================================
STEP 8 — Outcome Count
==================================================
```



•**Boxplots help in detecting outliers.**

```
==================================================
STEP 9 — Boxplots
==================================================
```



Expected Outcome

- **Identify key factors affecting diabetes.**
- **Provide useful insights for healthcare analysis.**
- **Improve data quality for future use.**
- **Support further machine learning model development.**

Chapter 4 CONCLUSION

4.1 Conclusion

This project successfully analyzes a real-world diabetes dataset using Pythonbased data science techniques. Various stages of data preprocessing, including handling missing values, removing outliers, and normalizing features, were performed to improve the overall quality, consistency, and reliability of the dataset. These preprocessing steps ensured that the data became clean, structured, and suitable for meaningful analysis.

Exploratory Data Analysis (EDA) and visualization techniques played a vital role in understanding the dataset. Graphical representations such as histograms, boxplots, countplots, and correlation heatmaps helped in identifying data distributions, class imbalance, and relationships among different medical attributes. Through these visualizations, important health indicators influencing diabetes, such as glucose level, BMI, age, and blood pressure, were clearly observed and analyzed.

The project demonstrates how data analysis can be effectively applied in the public health domain to extract valuable insights from healthcare data. It provides a strong analytical foundation for developing future machine learning models that can assist in diabetes prediction, early diagnosis, and preventive healthcare planning.

Overall, the project enhances understanding of healthcare analytics while highlighting the importance of data-driven decision-making in improving public health outcomes and awareness related to diabetes.